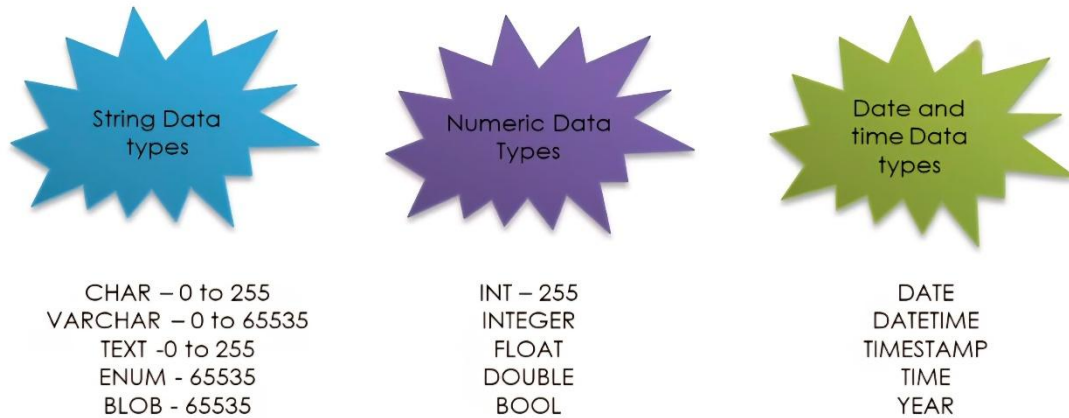


SQL (Structured Query Language)

➤ Data Types in SQL :-

SQL data types define the kind of data a column can hold. They are categorized into the following major types:



➤ Operators in SQL :-

- SQL Operators are symbols or keywords used to perform operations on data in a database.
- In SQL, operators are classified based on the type of operation they perform. Here are the main types of SQL operators.

(A) Comparison Operators :-

Comparison Operators	
=	a=b
!=	a!=b
>	a>b
<	a=	a>=b
<=	a<=b
!<	a!<b
!>	a!>b

(B) Arithmetic Operators :-

Arithmetic Operators

+	a+b
-	a-b
*	a*b
/	a/b
%	a%b

(C) Logical Operator :-

Logical Operators

ALL	compare a value to all values
AND	allows the existence of multiple conditions
ANY	compare the value in list according to the condition
BETWEEN	search for values, that are within a set of values
IN	compare a value to that specified list value
NOT	reverse the meaning of any logical operator
OR	combine multiple conditions
EXISTS	search for the presence of a row in a specified table
LIKE	compare a value to similar values using wildcard operator

➤ Operation on Database :-

(A) CREATE DATABASE

CREATE DATABASE

- The SQL **CREATE DATABASE** statement is used to create a new SQL database.

Syntax

CREATE DATABASE database_name;

Eg. **CREATE DATABASE** mydb;

Database name

(B) DROP DATABASE

DROP DATABASE

- The SQL **DROP DATABASE** statement is used to drop an existing database in SQL schema.

Syntax

DROP DATABASE DatabaseName;

Eg. **DROP DATABASE** mydb;

(C) RENAME DATABASE

RENAME DATABASE

- SQL **RENAME DATABASE** is used when you need to change the name of your database.

Syntax

RENAME DATABASE OldDatabaseName **TO** NewDatabaseName ;

Eg. **RENAME DATABASE** mydb **TO** mydb1;

Note: **RENAME DATABASE** Syntax was added in MySQL 5.1.7 but was found to be dangerous and was removed in MySQL 5.1.23.

(D) SELECT DATABASE

SELECT DATABASE

- When you have multiple databases in your SQL Schema, then before starting your operation, you would need to select a database

Syntax

USE Databasename;

Eg. **USE** mydb;

➤ Table in SQL :-

What is Table ?

- 1) Table is a collection of data, organized in terms of rows and columns.
- 2) In DBMS term, table is known as relation and row as tuple.
- 3) A table has a specified number of columns, but can have any number of rows.
- 4) A table is also considered as a convenient representation of relations.

Roll No	Name	Class	Marks	Age
1.	Anoop	10	30	22
2.	Anurag	10	32	23
3.	Ganesh	10	31	21

➤ Operation on Table :-

(A) CREATE TABLE

CREATE TABLE

- SQL **CREATE TABLE** statement is used to create table in a database.
- If you want to create a table, you should name the table and define its column and each column's data type.

Syntax

CREATE TABLE table_name(column_name1 datatype,
Column_name2 datatype,...);

Eg. **CREATE TABLE** student(id int(10), name varchar(20));

(B) DROP TABLE

DROP TABLE

- A SQL **DROP TABLE** statement is used to delete a table definition and all data from a table.

Syntax

DROP TABLE table_name;

Eg. **DROP TABLE** student;

(C) DELETE TABLE

DELETE TABLE

- The **DELETE** statement is used to delete rows from a table. If you want to remove a specific row from a table you should use **WHERE** condition.

Syntax

DELETE FROM table_name;

Eg. **DELETE FROM** student;

Syntax

DELETE FROM table_name [**WHERE** condition];

Eg. **DELETE FROM** student **WHERE** id=1;

(D) RENAME TABLE

RENAME TABLE

- SQL **RENAME TABLE** is used to change the name of a table.

Syntax

alter table old_table_name **rename to** new_table_name;

Eg. **alter table** student **rename to** student_new;

(E) TRUNCATE TABLE

TRUNCATE TABLE

- A truncate SQL statement is used to remove all rows (complete data) from a table.

Syntax

TRUNCATE TABLE table_name;

Eg. **TRUNCATE TABLE** student;

(F) COPY TABLE

COPY TABLE

- Copy a SQL table into another table in the same SQL server database

Syntax

insert into destination_table **select * from** source_table;

Eg. **Insert into** student1 **select * from** student;

Note: SELECT INTO is totally different from INSERT INTO statement.